Class-Conditional Batch Normalization Statistics Destroy Predictions but Preserve Features: Evidence That BN Running Statistics Act as Output Calibrators, Not Feature Encoders*

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Paper under double-blind review

Abstract

Batch Normalization (BN) running statistics are commonly understood as a benign normalization device whose only role at inference is to standardize activations. We test this assumption by replacing the global running statistics of trained convolutional networks with class-conditional statistics estimated from per-class subsets of the training data, and measuring two quantities: classification accuracy and the accuracy of a linear probe on the resulting penultimate features. Across four architectures (SmallResNet, VGG-11-BN, SimpleCNN, WideResNet-28-10) and three datasets (CIFAR-10, CIFAR-100, Tiny-ImageNet-200), we observe a consistent and counter-intuitive phenomenon: substituting the BN statistics of class c when evaluating class-c samples collapses test accuracy to near zero (3.5% on SmallResNet/CIFAR-10, 0.60% on CIFAR-100, 0.17% on Tiny-ImageNet-200), while a linear probe on the same features recovers $\geq$ 99.7% accuracy and reaches a perfect 100% on CIFAR-100 and Tiny-ImageNet-200. A continuous interpolation between class-conditional and global statistics yields a smooth monotonic recovery in classifier accuracy but a non-monotone dip in probe accuracy near α=0.6, indicating two distinct geometric regimes. A GroupNorm control eliminates the effect entirely (81.6% accuracy preserved), and temperature scaling reduces the resulting expected calibration error from 0.394 to 0.151 but recovers exactly 0% of the lost accuracy. Together these results indicate that BN running statistics are not part of the learned representation but a structural calibrator coupling the encoder to the classifier head.

1 Introduction

Batch normalization (Ioffe and Szegedy, 2015) is among the most widely adopted techniques in deep learning. During training, BN normalizes activations using mini-batch statistics; at inference, it substitutes exponentially averaged running statistics accumulated during training. The standard view treats these running statistics as a faithful, dataset-level approximation of the true population mean and variance: a regularization-and-stability device that affects optimization but not what the network represents.

This paper interrogates that view by asking a simple question: what happens if we replace the running BN statistics at inference with statistics estimated only from samples of one class? A naive prediction, drawing on the intuition that BN merely standardizes activations, would be that class-conditional statistics for class c should at worst be neutral, or perhaps mildly helpful, when evaluating images of class c. They are after all the most representative statistics for that class.

Our experiments contradict this prediction in dramatic fashion. On CIFAR-10 with a SmallResNet, same-class statistics drop accuracy from 91.1% to 3.5%—below the 10% chance

*Code and data: https://anonymous.4open.science/r/bn-calibration baseline. On CIFAR-100 it falls from 70.1% to 0.60% (chance: 1%); on Tiny-ImageNet-200 from 60.8% to 0.17% (chance: 0.5%). Yet linear probes trained on the resulting penultimate features recover 99.9%, 100.0%, and 100.0% accuracy respectively, demonstrating that the underlying features remain perfectly class-discriminative.

We argue that this dissociation between feature quality and predictive accuracy is most naturally explained by treating BN running statistics not as a normalization device but as an implicit output calibrator: the trained network's classifier head depends on a precise alignment between the running statistics and the linear weights that follow them, and any reasonably-scaled but wrongly-aligned statistics induce a destructive offset between features and decision boundary that linear re-fitting can completely undo.

Contributions.

- We formalize and run a controlled BN-statistic-replacement protocol with four conditions (global, same-class, wrong-class, random-class) and a linear-probe diagnostic that separates feature quality from output behavior.
- We show the same-class collapse is robust across four architectures (SmallResNet, VGG-11-BN, SimpleCNN, WideResNet-28-10) and three datasets (CIFAR-10, CIFAR-100, Tiny-ImageNet-200), with paired t-tests yielding $p < 10^{-4}$ in every BN-equipped condition.
- We trace the gap between feature quality and predictions through an 11-point interpolation between class-conditional and global statistics, showing a smooth monotone recovery in classifier accuracy and a non-monotone dip in probe quality near the midpoint.
- We rule out the alternative that the effect is intrinsic to the architecture by training a SmallResNet with GroupNorm (Wu and He, 2018): there are no running statistics to corrupt, and accuracy is preserved.
- We show that the failure is structural, not merely a temperature problem: temperature scaling (Guo et al., 2017) reduces same-class ECE from 0.394 to 0.151 but recovers exactly 0% of the lost accuracy, demonstrating that the logits are rotated, not merely sharpened.
- We show counter-intuitively that wrong-class and random-class statistics are less damaging than same-class statistics on every architecture, ruling out the simple "mismatched stats are bad" explanation.

These results have practical implications for transfer learning, domain adaptation, and normalization-free architecture design.

2 Related Work

Batch Normalization. Ioffe and Szegedy (2015) introduced BN to reduce "internal covariate shift." Santurkar et al. (2018) challenged this narrative, showing BN primarily smooths the optimization landscape; Bjorck et al. (2018) confirmed similar smoothing benefits. These works focus on BN's training-time role; we examine its inference-time function and show that BN running statistics are a structural component of the inference-time computation.

Expressive power of BN parameters. Frankle et al. (2020) demonstrated that training only BN affine parameters (γ, β) of a randomly initialized network achieves surprisingly high accuracy on CIFAR-10. This shows BN parameters carry significant information but does not distinguish the roles of learned affine parameters versus running statistics. Our results isolate the running statistics specifically.

BN for domain adaptation. Li et al. (2018) proposed Adaptive Batch Normalization (AdaBN), replacing BN running statistics with target-domain statistics for domain adaptation. This implicitly treats BN statistics as domain-specific calibrators. Our work provides direct experimental evidence for why this works: BN statistics modulate output scale and alignment, not feature content.

Test-time adaptation. Wang et al. (2021) and Schneider et al. (2020) update BN statistics on test data to improve robustness to distribution shift. Our calibration perspective explains why BN statistics are so influential at test time despite not being learned through backpropagation.

BN and adversarial robustness. Benz et al. (2021) showed adversarial vulnerability correlates with BN statistics. Our work suggests this may be because adversarial perturbations disrupt the calibration that BN statistics provide.

Linear probing as a feature-quality diagnostic. Linear probes (Alain and Bengio, 2016; Kornblith et al., 2019) test whether class structure is linearly accessible in a representation. We use them in a counterfactual role: if a network mispredicts but a linear probe on its features still classifies correctly, the failure is in the head, not the encoder.

Neural network calibration. Guo et al. (2017) demonstrated that modern neural networks are often poorly calibrated and introduced temperature scaling as a simple post-hoc calibration method. We use expected calibration error (ECE) and temperature scaling to characterize the nature of the accuracy collapse under BN-statistic perturbation, showing it is fundamentally different from standard miscalibration: temperature scaling halves ECE but cannot recover any lost accuracy.

Normalization alternatives. GroupNorm (Wu and He, 2018) computes statistics over feature-map groups within each example and stores no running averages. Layer-, instance-, and weight-norm variants similarly avoid persistent batch-level statistics. We use GroupNorm as a control: it does not have the mechanism that, in BN networks, leaks class-conditional information into a calibrator the head depends on. Recent normalization-free architectures (Brock et al., 2021) replace BN with alternative scaling strategies; our findings suggest the critical function being replaced is activation calibration.

3 Method

3.1 Architectures

We evaluate four architectures to test generality:

1. SmallResNet. A compact ResNet with channel widths $[32, 64, 128, 256]$ across four stages of two residual blocks each, containing 20 BatchNorm2d layers. This is our primary architecture (10 seeds on CIFAR-10). For Tiny-ImageNet-200 (64×64 inputs), we use a SmallResNet64 variant with five stages and channel widths $[32, 64, 128, 256, 512]$, adapted for the higher resolution with an initial 3×3 convolution (stride 1) and additional downsampling stages.
2. VGG-11-BN. A VGG-11 variant with BatchNorm after each convolutional layer (8 BN layers), providing an architecture without skip connections (3 seeds).
3. SimpleCNN. A plain 4-layer convolutional network with BatchNorm (4 BN layers), testing the effect in a minimal architecture (3 seeds).
4. WideResNet-28-10. A wide residual network with depth 28 and widen factor 10 (Zagoruyko and Komodakis, 2016), incorporating dropout (p=0.3) within residual blocks. This architecture is substantially wider and deeper than SmallResNet, with significantly more BN layers and parameters, testing the effect in a high-capacity network with regularization (3 seeds).

All models are trained on CIFAR-10 with standard augmentation (random crop with padding 4, random horizontal flip) using SGD with momentum 0.9, weight decay 5×10^{-4}, initial learning rate 0.1, and cosine annealing. SmallResNet trains for 20 epochs on CIFAR-10 (10 seeds); VGG-11-BN and SimpleCNN train for 15 epochs (3 seeds each); WideResNet-28-10 trains for 20 epochs (3 seeds). We additionally train SmallResNet on CIFAR-100 (25 epochs, 5 seeds) and SmallResNet64 on Tiny-ImageNet-200 (25 epochs, 3 seeds, 64×64 images, 200 classes) to test dataset generalization. WideResNet-28-10 uses standard single-precision training.

3.2 Datasets

We evaluate on three datasets of increasing complexity:

1. CIFAR-10. 10 classes, 32×32 images, 50K training / 10K test. Our primary benchmark.
2. CIFAR-100. 100 classes, 32×32 images, 50K training / 10K test. Tests scaling to many classes.
3. Tiny-ImageNet-200. 200 classes, 64×64 images, 100K training / 10K validation. Tests generalization to higher resolution, larger label spaces, and more complex visual content. Images are center-cropped to 64×64 for evaluation.

3.3 Computing Class-Conditional BN Statistics

Let f_θ be a trained CNN containing BN layers $\{B_\ell\}$ with running means μ_ℓ and running variances σ_ℓ^2. For each class c we compute a class-conditional estimate $(\mu_\ell^{(c)}, (\sigma_\ell^{(c)})^2)$ as follows: we reset the BN buffers, set the model to train() mode (which causes BN to update its running statistics with the standard exponential moving average), iterate the model in torch.no_grad() over the subset of the training set with label c, and snapshot the resulting buffers. The model parameters are not updated during this pass. We then return the BN buffers to their global values and use the snapshot only at inference. This yields C sets of BN statistics $\{\mu_c, \sigma_c^2\}$ for $c \in \{0, \dots, C-1\}$, one per class.

3.4 Evaluation Conditions

We evaluate under four BN-statistic conditions:

1. Global (baseline). Standard running statistics from training—normal inference.
2. Same-class. For test samples of class c, load $(\mu_\ell^{(c)}, (\sigma_\ell^{(c)})^2)$. Under the feature-encoding hypothesis, this should help or be neutral.
3. Wrong-class. For test samples of class c, load statistics from class $(c + \lfloor C/2 \rfloor) \bmod C$—a deliberately mismatched condition (e.g., class 0 receives class 5's statistics on CIFAR-10).
4. Random-class. For each test sample of class c, load statistics from a uniformly random different class.

For each condition we record per-class accuracy, per-class confidence (the softmax probability assigned to the true label), and the penultimate-layer features.

3.5 Metrics

For each condition:

- Classification accuracy. Top-1 accuracy using the trained classification head.
- Mean confidence. Average softmax probability assigned to the true class.
- Linear probe accuracy. We extract features from the penultimate layer (post global average pooling), train a linear classifier on an 80/20 split, and report held-out accuracy. We use ridge regression ($\lambda = 10^{-3}$, one-hot targets, closed form) for CIFAR-10, and multinomial logistic regression (L-BFGS, C=1.0, 1000 iterations) for $\geq$ 100-class settings (CIFAR-100, Tiny-ImageNet, WideResNet-CIFAR-100), where logistic regression is more stable.
- Expected calibration error (ECE). Following Guo et al. (2017), we compute ECE with 15 equal-width bins to quantify the gap between predicted confidence and actual accuracy.

Table 1: SmallResNet on CIFAR-10 (mean ± std, n=10 seeds).

Condition	Accuracy (%)	Confidence (%)	Linear Probe (%)
Global (baseline)	91.08 ± 0.12	88.88 ± 0.15	90.86 ± 0.55
Same-class	3.55 ± 0.69	4.80 ± 0.52	99.88 ± 0.09
Wrong-class	65.37 ± 4.69	50.64 ± 3.22	100.00 ± 0.00
Random-class	58.99 ± 5.84	45.47 ± 5.19	97.63 ± 1.11

3.6 Interpolation Experiment

To characterize the transition between same-class and global statistics, we linearly interpolate:

$$\mu_\alpha = \alpha \cdot \mu_{\text{global}} + (1-\alpha) \cdot \mu_{\text{same}} \tag{1}$$

$$\sigma^2_\alpha = \alpha \cdot \sigma^2_{\text{global}} + (1-\alpha) \cdot \sigma^2_{\text{same}} \tag{2}$$

for $\alpha \in \{0.0, 0.1, \ldots, 1.0\}$, evaluating classification accuracy and linear probe accuracy at each point.

3.7 GroupNorm Control

To confirm that the observed effect is specific to BN's running-statistic mechanism, we train a variant of SmallResNet with GroupNorm (Wu and He, 2018) replacing all BatchNorm layers (8 groups, or fewer when channels are smaller). GroupNorm normalizes using group statistics computed from the current input rather than maintained running statistics, so there are no class-conditional statistics to substitute. The relevant question is whether the architecture without BN suffers any analogous accuracy loss.

3.8 Expected Calibration Error and Temperature Scaling

To characterize the nature of the accuracy collapse beyond accuracy, we measure ECE (Guo et al., 2017) with 15 equal-width confidence bins. To test whether the accuracy collapse under same-class statistics is merely a logit-scale issue recoverable by post-hoc rescaling, we then apply temperature scaling (Guo et al., 2017): we optimize a single scalar temperature T on a held-out validation set to minimize negative log-likelihood under the same-class condition, then evaluate whether dividing logits by T recovers classification accuracy. Temperature scaling is a monotone transformation that preserves logit ordering (and thus the argmax), making it diagnostic: if accuracy improves, the issue is confidence scaling; if not, the logit ranking itself is disrupted.

3.9 Statistical Protocol

SmallResNet on CIFAR-10 uses 10 independent seeds (42–51). SmallResNet on CIFAR-100 uses 5 seeds (42–46). All other configurations use 3 seeds (42–44). We report means ± standard deviations and conduct paired t-tests for key comparisons.

4 Results

4.1 Primary Finding: SmallResNet on CIFAR-10

Table 1 presents aggregate results across 10 seeds.

The results reveal a dramatic dissociation:

1. Same-class BN statistics catastrophically degrade accuracy to 3.5%, well below the 10% chance baseline. Providing the network with class-appropriate statistics is actively destructive.

Table 2: Cross-architecture results on CIFAR-10 (mean ± std, n=3 seeds).

Architecture	Condition	Acc. (%)	Conf. (%)	Probe (%)
VGG-11-BN	Global	85.44 ± 1.28	80.47 ± 1.78	85.00 ± 1.80
	Same-class	10.88 ± 1.91	10.29 ± 1.23	98.37 ± 0.63
	Wrong-class	62.96 ± 1.25	58.61 ± 1.31	99.57 ± 0.27
	Random-class	57.50 ± 3.03	52.29 ± 3.12	95.28 ± 0.78
SimpleCNN	Global	83.17 ± 0.23	75.68 ± 0.11	80.53 ± 1.44
	Same-class	26.33 ± 0.48	20.71 ± 0.18	54.68 ± 0.74
	Wrong-class	78.10 ± 0.46	65.79 ± 0.59	93.67 ± 0.27
	Random-class	72.11 ± 0.36	59.20 ± 0.33	90.82 ± 0.52
WRN-28-10	Global	90.61 ± 0.51	89.36 ± 0.43	91.02 ± 0.28
	Same-class	11.12 ± 0.90	11.04 ± 0.73	99.78 ± 0.09
	Wrong-class	64.06 ± 3.14	52.64 ± 3.51	99.60 ± 0.07
	Random-class	62.41 ± 1.76	51.21 ± 2.25	96.63 ± 0.44

2. Wrong-class statistics outperform same-class (65.4% vs. 3.5%), a complete inversion of the feature-encoding prediction.

3. Linear probe accuracy under same-class statistics reaches 99.9%, far exceeding the global baseline (90.9%). Representations become more linearly separable under perturbation.

4. Statistical significance. Same vs. global accuracy: $t(9) = -374.1$, $p = 3.5 \times 10^{-20}$. Same vs. global linear probe: $t(9) = +48.7$, $p = 3.3 \times 10^{-12}$.

Per-class deltas (same-class minus global) are uniformly large and negative across all 10 CIFAR-10 classes, ranging from -0.97 for automobile to -0.72 for cat (the smallest absolute drop): the collapse is not driven by one or two pathological classes.

4.2 Cross-Architecture Replication

Table 2 shows the effect replicates across architecturally distinct networks.

All four architectures exhibit the same core pattern: same-class statistics devastate classification accuracy while linear probe accuracy remains high or increases.

VGG-11-BN shows the same pattern: same-class statistics reduce accuracy from 85.4% to 10.9% while linear probe accuracy increases from 85.0% to 98.4%. SimpleCNN shows the accuracy collapse (83.2% → 26.3%) but with reduced linear probe improvement (80.5% → 54.7%), likely reflecting its simpler representational capacity with only 4 BN layers and a small (256-dim) feature space.

WideResNet-28-10 provides particularly strong evidence because it is a high-capacity architecture with dropout regularization (p=0.3). Despite achieving the highest baseline accuracy (90.6%), same-class statistics reduce accuracy to 11.1% ($t(2) = -95.94$, $p = 0.00011$) while the linear probe reaches 99.8% ($t(2) = +55.14$, $p = 0.00033$). The wrong-class condition (64.1%) again substantially outperforms same-class (11.1%), with $t(2) = -30.64$, $p = 0.0011$. Notably, WideResNet's same-class accuracy (11.1%) is higher than SmallResNet's (3.5%), which may reflect the effect of dropout: by adding noise during training, dropout may make the classification head slightly more robust to the activation perturbation induced by BN-statistic replacement.

The severity of the accuracy collapse correlates with the number of BN layers: SmallResNet (20 BN layers) drops to 3.5%, VGG-11-BN (8 layers) to 10.9%, WideResNet-28-10 (many BN layers, but with dropout) to 11.1%, and SimpleCNN (4 layers) to 26.3%. More BN layers generally means more opportunities for calibration disruption to compound, though dropout may partially mitigate the compounding effect.

Table 3: Interpolation from same-class (α=0) to global (α=1) statistics.

α	Accuracy (%)	Linear Probe (%)
0.0	3.44	99.90
0.1	7.13	99.70
0.2	13.41	98.67
0.3	23.14	94.85
0.4	36.47	87.10
0.5	51.61	77.60
0.6	65.17	74.65
0.7	75.26	77.88
0.8	82.62	83.00
0.9	87.64	87.53
1.0	91.01	90.85

Table 4: SmallResNet on CIFAR-100 (mean $\pm$ std, n=5 seeds). Linear probe uses logistic regression.

Condition	Acc. (%)	Conf. (%)	Probe (%)
Global	70.11 ± 0.57	63.98 ± 0.73	64.23 ± 1.34
Same-class	0.60 ± 0.49	0.94 ± 0.00	100.00 ± 0.00
Wrong-class	1.00 ± 0.00	1.00 ± 0.00	100.00 ± 0.00
Random-class	1.00 ± 0.63	1.00 ± 0.00	65.49 ± 8.75

4.3 Interpolation Between Same-Class and Global Statistics

Table 3 shows the interpolation results (SmallResNet, CIFAR-10).

As α increases from 0 to 1, classification accuracy rises smoothly and monotonically from 3.4% to 91.0%—there is no phase transition. Linear probe accuracy, however, is non-monotone: it is highest at the same-class endpoint (99.9%), declines to a minimum of 74.65% at α=0.6, and then partially recovers to 90.85% at the global endpoint. The two metrics briefly cross near $\alpha\approx$0.6–0.7, where classifier accuracy and probe accuracy approach equality.

This anti-correlation in the lower half of the range is the signature of the calibration mechanism: the more calibrated the BN statistics are to the head's expectations, the higher classification accuracy but the lower probe accuracy. The non-monotone probe dip at α=0.6 further indicates that intermediate statistics simultaneously reduce the head's bias and the feature space's class-collapsed separability, producing a regime that is neither the "perfectly separable, head-misaligned" state at α=0 nor the "head-aligned, normally separable" state at α=1.

4.4 CIFAR-100 Generalization

To test whether the finding generalizes beyond 10-class problems, we repeat the experiment on CIFAR-100 with SmallResNet (5 seeds, 25 epochs).

The accuracy collapse is even more severe on CIFAR-100: same-class statistics reduce accuracy from 70.1% to 0.60%, well below the 1% chance level for 100 classes ($t(4) = -162.9$, $p = 8.5 \times 10^{-9}$). All non-global conditions produce near-chance accuracy and near-uniform confidence ($\sim$1%), consistent with complete calibration failure across 100 output dimensions.

Linear probe results. With logistic regression probes, same-class and wrong-class conditions both achieve perfect 100% probe accuracy with zero variance across all 5 seeds—confirming that representations remain fully intact despite complete classification collapse.

Table 5: SmallResNet64 on Tiny-ImageNet-200 (mean ± std, n=3 seeds).

Condition	Acc. (%)	Conf. (%)	Probe (%)
Global	60.77 ± 0.30	53.75 ± 0.24	51.83 ± 0.55
Same-class	0.17 ± 0.24	0.49 ± 0.00	100.00 ± 0.00
Wrong-class	0.50 ± 0.00	0.50 ± 0.00	100.00 ± 0.00
Random-class	0.50 ± 0.00	0.50 ± 0.00	57.02 ± 3.41

Table 6: GroupNorm control on CIFAR-10 (mean ± std, n=3 seeds).

Metric	Value
Accuracy	$81.65 \pm 1.94\%$
Linear Probe	$81.08 \pm 1.64\%$

4.5 Tiny-ImageNet-200 Generalization

To test whether our findings extend to higher-resolution images, larger label spaces, and more complex visual content, we evaluate on Tiny-ImageNet-200 using SmallResNet64—a five-stage variant of SmallResNet adapted for 64×64 inputs (3 seeds, 25 epochs).

Tiny-ImageNet-200 produces the most dramatic results in our study. Same-class statistics reduce classification accuracy from 60.8% to 0.17%—well below the 0.5% chance level for 200 classes ($t(2) = -402.16$, $p = 6.2 \times 10^{-6}$). The network's confidence collapses to a uniform 0.49% (near the 0.5% expected under random guessing over 200 classes), indicating complete calibration failure. Wrong-class statistics similarly collapse accuracy to 0.50% ($t(2) = -280.89$, $p = 1.3 \times 10^{-5}$).

The linear probe results are striking: under both same-class and wrong-class conditions, the probe achieves perfect 100.0% accuracy with zero variance across seeds. This is the cleanest dissociation in our study—complete output collapse coexisting with perfectly separable representations. The 512-dimensional feature space of SmallResNet64 supports stable probe performance even for 200 classes.

The same-class vs. wrong-class comparison is also informative: on Tiny-ImageNet, same-class accuracy (0.17%) is nominally lower than wrong-class (0.50%), though this difference does not reach statistical significance ($t(2) = -2.0$, $p = 0.18$)—both are effectively at floor. With 200 classes, any BN-statistic perturbation causes total calibration failure, leaving no room for the same-class vs. wrong-class distinction that is visible with 10 classes. The random-class probe is lower (57.0%) because each sample sees a different class's statistics, averaging over inconsistent perturbation directions. The size of the probe-vs-prediction gap closely tracks the number of classes, consistent with the structural-calibration interpretation: a fixed misalignment of statistics in a K-way head dominates a K-way argmax more decisively as K grows.

4.6 GroupNorm Control

To confirm the effect is specific to BN's running-statistic mechanism, we train SmallResNet with GroupNorm replacing all BatchNorm layers (3 seeds, 20 epochs).

GroupNorm models achieve 81.6% accuracy—comparable to the BatchNorm baseline (91.1%) and within the expected range for this architecture without BN's optimization benefits. Crucially, since GroupNorm computes normalization statistics from the current input at test time, there are no stored running statistics to manipulate. The class-conditional replacement paradigm is inapplicable, confirming that the accuracy collapse observed in Sections 4.1–4.5 is specific to BatchNorm's running-statistic mechanism, not a general property of normalization layers.

Table 7: Expected Calibration Error by condition (SmallResNet, CIFAR-10, n=3 seeds).

Condition	Accuracy (%)	ECE
Global (baseline)	91.33 ± 0.12	0.025 ± 0.001
Same-class	2.87 ± 0.49	0.394 ± 0.013
Wrong-class	63.40 ± 4.26	0.064 ± 0.013

Table 8: Temperature scaling on same-class condition (SmallResNet, CIFAR-10, n=3 seeds).

Metric	Before	After ($T \approx 3.43$)
Accuracy (%)	2.87 ± 0.49	2.87 ± 0.49
ECE	0.394 ± 0.013	0.151 ± 0.003

4.7 Calibration Analysis

To characterize the nature of the accuracy collapse, we measure expected calibration error (ECE; Guo et al., 2017) across BN-statistic conditions on SmallResNet CIFAR-10 (3 seeds).

Under global statistics, the network is well-calibrated (ECE = 0.025). Same-class statistics induce severe miscalibration (ECE = 0.394): the network assigns confident predictions that are systematically wrong. Interestingly, wrong-class statistics produce moderate ECE (0.064) despite substantially reduced accuracy—the network's confidence and accuracy remain roughly aligned, just at a lower level. This pattern is consistent with the calibration hypothesis: same-class statistics specifically disrupt the mapping between activation magnitudes and output confidence, rather than destroying the underlying feature representation.

4.8 Temperature Scaling Cannot Recover Accuracy

A natural question is whether the accuracy collapse under same-class statistics is merely a logit-scale issue—perhaps the correct class still receives the highest pre-softmax score, but the softmax temperature is wrong. If so, temperature scaling should recover accuracy.

Temperature scaling finds an optimal temperature of $T \approx 3.43 \pm 0.02$, indicating that logits under same-class statistics are substantially over-scaled. The high temperature successfully reduces ECE from 0.394 to 0.151—a meaningful improvement in calibration. However, accuracy remains exactly unchanged at 2.87%, recovering 0% of the lost classification performance. This is by construction (temperature scaling preserves the argmax) but the size of the residual ECE makes clear that the underlying logit structure is not merely too sharp—it is rotated.

This result rules out a simple explanation of our findings. The accuracy collapse is not merely an issue of logit magnitude or softmax sharpness. Rather, same-class BN statistics fundamentally alter which class receives the highest logit, scrambling the output layer's class assignments. The network's predictions are not just poorly calibrated—they are systematically wrong in a way that no post-hoc logit rescaling can fix. This confirms that BN statistics control a structural calibration that aligns internal activation patterns with the classification head's learned decision boundaries, not merely the scale of the output distribution.

5 Discussion

5.1 Converging Lines of Evidence

Our results constitute eight independent lines of evidence that BN running statistics function as output calibrators rather than feature encoders:

1. Same-class statistics destroy accuracy but preserve representations (CIFAR-10, 4 architectures). The dramatic dissociation between 3.5% classification accuracy and 99.9% linear probe accuracy is inexplicable under the feature-encoding hypothesis.
2. The effect is amplified with more classes. On CIFAR-100, same-class statistics reduce accuracy to 0.60%—below the 1% chance level. On Tiny-ImageNet-200, accuracy drops to 0.17%—below the 0.5% chance level—demonstrating that the calibration disruption scales with output-space dimensionality.
3. The effect generalizes to higher resolution. On Tiny-ImageNet-200 (64×64 images, 200 classes), the dissociation is the most extreme observed: 0.17% classification accuracy with perfect 100% linear probe accuracy across all seeds. This rules out the possibility that the effect is an artifact of low-resolution inputs.
4. The effect is BN-specific. GroupNorm models, which lack running statistics, are immune by construction (81.6% accuracy preserved).
5. Smooth interpolation, but a non-monotone probe dip. The smooth monotonic transition in classification accuracy between same-class and global statistics rules out discrete phase transitions, while the non-monotone dip in probe accuracy at $\alpha \approx 0.6$ shows that the two endpoints correspond to two distinct geometric regimes for the feature space.
6. ECE analysis reveals structured miscalibration. Same-class statistics produce severely miscalibrated outputs (ECE = 0.394) while wrong-class statistics remain surprisingly well-calibrated (ECE = 0.064), indicating that the disruption is specifically in how activations are scaled for downstream interpretation.
7. Temperature scaling cannot recover accuracy. Post-hoc calibration reduces ECE (0.394 → 0.151) but yields zero accuracy recovery (2.9% → 2.9%), proving that same-class statistics rotate logit rankings, not merely sharpen them.
8. Dropout does not prevent the effect. WideResNet-28-10, which uses dropout (p=0.3) as additional regularization, still shows catastrophic accuracy collapse (90.6% → 11.1%) with near-perfect probe accuracy (99.8%), demonstrating that the calibration dependence is not an artifact of overfitting to specific activation scales.

5.2 Why Same-Class Statistics Are Worse Than Wrong-Class

The most counterintuitive finding is that same-class BN statistics (3.5%) perform worse than wrong-class (65.4%) or random-class (59.0%) statistics. We propose the following explanation.

The classification head of a BN-trained network is trained to apply a specific affine transformation after normalization with the global statistics; that transformation effectively learns per-feature offsets and scales that bake in the global running statistics. Substituting class-c statistics shifts the feature distribution away from the head's expected input in the direction the head is most sensitive to for class c, because those statistics encode the class-mean activation pattern. Wrong- and random-class statistics shift it in less specifically targeted directions, where the head's misalignment is smaller. Linear refitting (the probe) absorbs whatever offset the substituted statistics cause, which is why probe accuracy is uniformly high.

A complementary perspective is that same-class statistics normalize away the between-class activation variation that the final linear layer exploits. Each class gets "centered" onto itself, collapsing the between-class signal. Wrong-class statistics introduce a mismatch, but one that may preserve or even amplify some between-class structure.

This is consistent with the linear-probe results: same-class normalization makes representations more linearly separable (99.9% probe accuracy) by removing within-class variation, but the original classification head cannot exploit this because it was calibrated for globally-normalized inputs.

The ECE analysis (Section 4.7) adds nuance: wrong-class statistics produce well-calibrated but less accurate outputs (ECE = 0.064, accuracy = 63.4%), suggesting the mismatch

Table 9: Architecture-dependent sensitivity to same-class BN-statistic replacement.

Architecture	BN Layers	Same-Class Acc. (%)	Notes
SmallResNet	20	3.55	Deepest collapse
VGG-11-BN	8	10.88	No skip connections
WideResNet-28-10	Many	11.12	Has dropout (p=0.3)
SimpleCNN	4	26.33	Fewest BN layers

introduces noise but preserves the approximate relationship between confidence and correctness. Same-class statistics produce severely miscalibrated outputs (ECE = 0.394, accuracy = 2.9%), indicating a systematic rather than random disruption—the network is confidently wrong, not merely uncertain.

On Tiny-ImageNet-200, this distinction collapses: with 200 classes, both same-class (0.17%) and wrong-class (0.50%) produce floor-level accuracy, as the calibration disruption is severe enough to completely scramble any residual class structure in the outputs.

5.3 The Calibration Decomposition

Our results support a clean functional decomposition of BN-equipped networks:

1. Convolutional filters learn class-discriminative features that are robust to normalization perturbation.
2. BN running statistics calibrate activation scale and shift to match downstream expectations.
3. The classification head is a thin linear layer highly sensitive to the specific calibration of its inputs.

The interpolation experiment (Section 4.3) makes this vivid: classification accuracy and linear probe accuracy move in opposite directions over the lower half of the interpolation range, and the probe dips to a minimum near α=0.6 before partially recovering. The optimal calibration for the classification head (α=1, global) is not the optimal point for linear separability of the underlying features.

The calibration analysis (Sections 4.7–4.8) provides further mechanistic evidence. The high ECE under same-class statistics (0.394) shows the network is confidently wrong—not uncertain, but miscalibrated. Temperature scaling reduces this miscalibration (ECE: 0.394 $\rightarrow$ 0.151) without recovering accuracy, demonstrating that the disruption operates at the level of class-identity assignment, not output confidence scaling. BN statistics control which activation patterns map to which classes, not merely how confident the mapping is.

The Tiny-ImageNet-200 results (Section 4.5) provide the most compelling evidence for this decomposition. The five-stage SmallResNet64 learns rich 512-dimensional representations that remain perfectly linearly separable under BN-statistic perturbation—even as the classification head produces completely random outputs. This perfect dissociation, achieved with 200 classes at 64×64 resolution, demonstrates that the calibration decomposition holds across scales.

5.4 Architecture-Dependent Sensitivity

The severity of the accuracy collapse under same-class statistics shows interesting variation across architectures that is broadly (but not perfectly) correlated with the number of BN layers:

SmallResNet, with 20 BN layers and no dropout, shows the most severe collapse (3.5%). SimpleCNN, with only 4 BN layers, retains the most accuracy (26.3%), consistent with fewer calibration-disruption points. WideResNet-28-10 is an interesting case: despite having many BN layers and a very high-capacity architecture, its same-class accuracy (11.1%) is notably higher than SmallResNet's (3.5%). This may be attributable to dropout (p=0.3), which introduces stochastic noise during training and may make the classification head slightly more robust to the kind of activation perturbation induced by BN-statistic replacement. Dropout effectively trains the classification head to tolerate some activation variance, partially buffering it against the calibration disruption.

5.5 Implications

Transfer learning. Resetting BN running statistics on a new domain is effective because it recalibrates activation scales, not because it encodes new features. Our results predict that fine-tuning the classification head after BN-statistic reset should be sufficient for adaptation, even without retraining convolutional weights.

Domain adaptation. The success of AdaBN (Li et al., 2018)—replacing BN statistics with target-domain statistics—follows naturally from the calibration perspective: target-domain statistics recalibrate activations to a consistent scale the classification head can interpret.

Test-time adaptation. Methods that update BN statistics on test data (Wang et al., 2021; Schneider et al., 2020) interact with the classifier head in ways the literature has not fully characterized. Our results suggest such methods should be paired with classifier-head re-alignment to fully exploit the recalibrated activations.

Normalization-free architectures. Recent architectures like NFNets (Brock et al., 2021) replace BN with alternative scaling strategies. Our findings suggest the critical function being replaced is activation calibration. Simpler alternatives focused specifically on maintaining consistent activation scales may suffice.

Model compression. BN folding (absorbing BN into convolutional weights during deployment) bakes a specific calibration into the weights. Post-hoc weight modifications must preserve this calibration, not just feature-extraction capability.

Post-hoc calibration. Our temperature-scaling results (Section 4.8) reveal a fundamental distinction between output calibration (adjusting confidence to match accuracy, as in Guo et al. 2017) and structural calibration (aligning internal activation scales with downstream expectations). BN statistics perform structural calibration; temperature scaling performs output calibration. The two are complementary but address different failure modes.

5.6 Broader Significance

The methodological lesson is perhaps most important: classification accuracy can be deeply misleading about representation quality. A model at 3.5% accuracy—worse than random chance—may have learned excellent class-discriminative features. On Tiny-ImageNet-200, a model at 0.17% accuracy—a third of the chance rate—achieves perfect linear probe accuracy. The bottleneck is not representation but calibration. This suggests that many apparent failures in deep learning (e.g., in domain shift, adversarial settings, or after architectural modifications) may be calibration failures rather than representation failures.

The ECE analysis reinforces this point quantitatively: the same-class condition produces a network that is both severely inaccurate and severely miscalibrated, yet its internal representations are near-perfectly separable. Future work on diagnosing model failures should consider separately measuring representation quality and output calibration rather than relying on end-to-end accuracy alone.

6 Limitations

1. Architecture and scale. While we test four architectures spanning residual networks (SmallResNet, WideResNet-28-10), plain convnets (VGG-11-BN), and minimal networks (SimpleCNN), all are trained on relatively small-to-moderate-scale datasets. Scaling to ImageNet-1k, models with hundreds of layers, and transformer architectures with non-BN normalization is left to future work.
2. Dataset scope. CIFAR-10, CIFAR-100, and Tiny-ImageNet-200 span 10 to 200 classes and 32×32 to 64×64 resolution, but remain within the domain of natural-

image classification. Fine-grained recognition, medical imaging, or natural language tasks may show different dynamics.

3. Training duration. Models are trained for 15–25 epochs on Apple Silicon (MPS backend). Longer training schedules may yield different BN-statistic structure.

4. Linear probe limitations. Ridge regression and logistic regression test only linear separability; we use logistic regression for $\geq$ 100-class probes for stability. Non-linear probe families could reveal additional structure but would also complicate the "head misalignment" interpretation.

5. Affine vs. running-statistic decoupling. BN computes $\gamma \cdot (x - \mu)/\sigma + \beta$. When we replace the running statistics (μ, σ) we leave the affine parameters (γ, β) unchanged, but the effective activation distribution after the layer shifts as a function of both. We did not explicitly isolate the contribution of the affine parameters from that of the running statistics. Future work could disentangle these by also varying (γ, β).

6. Wrong-class mapping is not exhaustive. We use a fixed rule for the wrong-class condition (class $c \to (c + \lfloor C/2 \rfloor) \bmod C$). A richer analysis would consider all C^2 stat-class pairings and characterize the geometry of the resulting accuracy surface.

7. Static replacement. We replace all BN layers simultaneously. Layer-wise replacement could reveal which layers are most sensitive to calibration disruption, and whether the effect compounds linearly or multiplicatively across the network depth.

8. ECE limitations. ECE with fixed binning can be sensitive to bin count and sample size (Nixon et al., 2019). We use 15 bins following standard practice, but kernel-based calibration measures could provide smoother estimates.

9. Architecture-dependent feature robustness. The SimpleCNN result (where the same-class probe drops to 54.7% rather than rising) hints at architecture-dependent feature robustness that we have not characterized systematically. Probing networks with different feature dimensionalities and depths is a natural follow-up.

7 Conclusion

We demonstrate a striking dissociation between classification accuracy and representation quality under batch normalization statistic manipulation. Replacing global BN statistics with same-class statistics destroys classification (3.5% on CIFAR-10, 0.60% on CIFAR-100, 0.17% on Tiny-ImageNet-200) while yielding near-perfect or perfect linear probe accuracy (99.9% on CIFAR-10, 100.0% on CIFAR-100, 100.0% on Tiny-ImageNet-200). This effect replicates across four architectures—including WideResNet-28-10 with dropout—generalizes across three datasets spanning 10 to 200 classes and 32×32 to 64×64 resolution, varies smoothly under interpolation (with a non-monotone probe dip at the midpoint), and is specific to BatchNorm's running-statistic mechanism (absent with GroupNorm).

Calibration analysis reveals that same-class statistics induce severe miscalibration (ECE = 0.394 vs. 0.025 baseline), and temperature scaling—while halving ECE—recovers exactly 0% of the lost accuracy, confirming that the collapse reflects structural rotation of logits rather than a simple sharpening or rescaling. These findings establish that BN running statistics function as output calibrators rather than feature encoders. Networks learn representations that are remarkably robust to BN perturbation, but classification heads are precisely calibrated to the activation scale produced by the global statistics, and they should be analyzed—and intervened upon—with the same care normally given to learned parameters.

This calibration perspective unifies observations from transfer learning, domain adaptation, and test-time adaptation, and has implications for the design of normalization layers and calibration-aware training procedures.

Reproducibility. Code and data are available at https://anonymous.4open.science/r/bn-calibration.

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A Experimental Configuration

B Full Per-Seed Results

Table 10: Full experimental configuration.

Parameter	Value
SmallResNet	BasicBlock, $[2, 2, 2, 2]$, channels $[32, 64, 128, 256]$, 20 BN layers
SmallResNet64	BasicBlock, 5-stage, channels $[32, 64, 128, 256, 512]$, for 64×64
VGG-11-BN	Standard VGG-11 with BN, 8 BN layers
SimpleCNN	4-layer CNN, 4 BN layers
WideResNet-28-10	depth=28, widen_factor=10, dropout=0.3
Training epochs	20 (CIFAR-10), 25 (CIFAR-100, Tiny-ImageNet-200)
Batch size	128
Optimizer	SGD (momentum=0.9, weight_decay=5×10^{-4})
Learning rate	0.1, cosine annealing
Augmentation	RandomCrop(32, pad=4), RandomHorizontalFlip (CIFAR)
Normalization	μ=$(0.4914, 0.4822, 0.4465)$, σ=$(0.2470, 0.2435, 0.2616)$
Linear probe	Ridge (λ=10^{-3}) or LogisticRegression, 80/20 split
ECE	15 equal-width bins
Temp. scaling	NLL-optimized scalar T on validation set
Seeds	42–51 (SmallResNet CIFAR-10), 42–46 (SmallResNet CIFAR-100), 42–44 (all others)
Device	Apple Silicon (MPS backend)

Table 11: SmallResNet CIFAR-10 — Classification accuracy by seed and condition.

Seed	Global	Same-Class	Wrong-Class	Random-Class
42	90.96	3.85	60.19	55.26
43	91.05	4.02	70.70	67.73
44	91.01	2.46	69.93	55.45
45	91.07	4.06	64.59	56.74
46	91.26	3.84	57.08	56.07
47	91.30	3.37	60.49	57.01
48	90.95	2.39	67.19	50.82
49	91.19	2.90	63.57	56.01
50	90.93	4.22	69.00	66.09
51	91.04	4.35	70.98	68.72

Table 12: SmallResNet CIFAR-10 — Linear probe accuracy by seed and condition.

Seed	Global	Same-Class	Wrong-Class	Random-Class
42	90.30	99.90	100.00	96.30
43	90.65	99.80	100.00	96.85
44	91.10	99.90	100.00	97.05
45	89.65	99.90	100.00	96.90
46	90.80	100.00	100.00	98.30
47	91.80	99.95	100.00	99.35
48	90.80	99.90	100.00	96.80
49	91.15	99.90	100.00	98.90
50	91.00	99.65	100.00	96.65
51	91.30	99.90	100.00	99.15

Table 13: SmallResNet CIFAR-100 — Per-seed results (n=5 seeds, logistic regression probe).

Seed	Glob. Acc	SC Acc	Glob. Probe	SC Probe
42	70.95	1.00	65.65	100.00
43	69.68	1.00	64.85	100.00
44	70.21	0.00	65.00	100.00
45	69.31	1.00	61.80	100.00
46	70.42	0.00	63.85	100.00

Table 14: ECE and temperature scaling — Per-seed results (SmallResNet, CIFAR-10).

Seed	Glob. ECE	SC ECE	SC ECE (post-T)	T
42	0.026	0.382	0.150	3.456
43	0.024	0.412	0.148	3.428
44	0.026	0.388	0.154	3.416

Table 15: WideResNet-28-10 CIFAR-10 — Per-seed classification accuracy (n=3 seeds).

Seed	Global Acc	Same-Class Acc	Wrong-Class Acc	Random-Class Acc
42	90.17	10.44	64.21	62.49
43	90.35	12.40	67.83	64.53
44	91.32	10.53	60.15	60.22

Table 16: WideResNet-28-10 CIFAR-10 — Per-seed linear probe accuracy (n=3 seeds).

Seed	Global Probe	Same-Class Probe	Wrong-Class Probe	Random-Class Probe
42	91.40	99.85	99.70	96.30
43	90.90	99.85	99.55	96.35
44	90.75	99.65	99.55	97.25

Table 17: SmallResNet64 Tiny-ImageNet-200 — Per-seed results (n=3 seeds).

Seed	Glob. Acc	Glob. Conf	Glob. Probe	SC Acc	SC Probe
42	60.88	53.78	51.55	0.00	100.00
43	61.08	54.04	52.60	0.50	100.00
44	60.36	53.44	51.35	0.00	100.00